WORKED SOLUTION

STEP III 2007, Question 9

Mechanics • Simple Harmonic Motion

STEP III 2007, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q9

The key idea

The spring pulls or pushes the two beads equally and oppositely along the same line, and the hoop's reactions are radial, so the total angular momentum about the centre stays zero: the beads move as mirror images. That reduces everything to one coordinate, and conservation of energy then gives simple harmonic motion in θ. The catch is that θ is the *acute* half-angle: if the beads get far enough apart to pass through opposite ends of a diameter, the formula for the energy changes form, and the motion must be pieced together from two arcs of SHM.

Solution

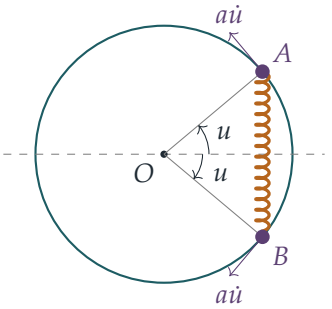
Setting up

Model the beads as particles on a fixed, smooth hoop, and the spring as light. The forces on the beads are gravity and the vertical part of the hoop's reaction (which cancel), the horizontal part of the reaction (along the radius, since the hoop is smooth) and the spring forces. The spring force on A and the spring force on B are equal and opposite and act along the same line AB, so their total moment about the vertical axis through O is zero; the radial forces have no moment about it at all. Hence the total angular momentum about that axis is conserved, and it is zero because the beads start at rest.

Measure angles round the hoop from a fixed radius, so that A and B are at angles ϕ_A and ϕ_B. Zero angular momentum means $ma^2(\dot{\phi}_A + \dot{\phi}_B) = 0$, so $\phi_A + \phi_B$ is constant. Choose the reference radius to bisect the initial angle AOB; then for all time

$$\phi_A = u, \qquad \phi_B = -u,$$

for a single coordinate u, with $u = \beta$ at the start. Each bead has speed $a\dot{u}$, so the kinetic energy is $2 \cdot \frac{1}{2}ma^2\dot{u}^2 = ma^2\dot{u}^2$.



The angle AOB is $2u$ while $0 < u \leq \frac{\pi}{2}$, but if u exceeds $\frac{\pi}{2}$ the beads have passed opposite ends of a diameter and the angle AOB is $2\pi - 2u$. So

$$\theta = u \quad (0 < u \leq \tfrac{\pi}{2}), \qquad \theta = \pi - u \quad (\tfrac{\pi}{2} \leq u < \pi).$$

Conservation of energy, with the beads released from rest at $u = \beta$, gives

$$ma^2\dot{u}^2 + mk^2a^2(\theta - \alpha)^2 = mk^2a^2(\beta - \alpha)^2, \qquad \text{so} \qquad \dot{u}^2 = k^2[(\alpha - \beta)^2 - (\theta - \alpha)^2].$$

"In compression" means the spring is shorter than its natural length. The energy is least at $\theta = \alpha$, so that is the natural length, and the length $2a\sin\theta$ increases with θ on $(0, \frac{\pi}{2})$; hence $0 < \beta < \alpha$.

The motion while $u \leq \frac{\pi}{2}$

Here $\theta = u$, and differentiating the energy equation with respect to t and dividing by $2\dot{u}$ (valid whenever the beads are moving, and then by continuity) gives

$$\ddot{\theta} = -k^2(\theta - \alpha).$$

This is SHM about $\theta = \alpha$ with angular frequency k. Starting from rest at $\theta = \beta$,

$$\theta - \alpha = -(\alpha - \beta)\cos kt,$$

so the beads move apart and, if nothing interrupts them, reach $\theta = 2\alpha - \beta$ at $t = \frac{\pi}{k}$. Everything depends on whether $2\alpha - \beta$ is below $\frac{\pi}{2}$, that is, on how β compares with $2\alpha - \frac{\pi}{2}$. This critical value lies in $(0, \alpha)$ precisely because $\frac{\pi}{4} < \alpha < \frac{\pi}{2}$, so all three cases below can occur.

Case 1: $2\alpha - \frac{\pi}{2} < \beta < \alpha$

Now $2\alpha - \beta < \frac{\pi}{2}$, so the beads never reach the ends of a diameter, $\theta = u$ throughout, and the motion is pure SHM between $\theta = \beta$ and $\theta = 2\alpha - \beta$. The period is

$$T = \frac{2\pi}{k}.$$

Case 2: $0 < \beta < 2\alpha - \frac{\pi}{2}$

Now θ reaches $\frac{\pi}{2}$ before the SHM turns round, at the time t_1 with

$$\tfrac{\pi}{2} - \alpha = -(\alpha - \beta)\cos kt_1, \qquad \text{so} \qquad \cos kt_1 = -\lambda, \quad \text{where } \lambda = \frac{\frac{\pi}{2} - \alpha}{\alpha - \beta}.$$

Since $\alpha < \frac{\pi}{2}$ and $\beta < 2\alpha - \frac{\pi}{2}$, we have $0 < \lambda < 1$. Hence

$$kt_1 = \pi - \arccos\lambda = \tfrac{\pi}{2} + \arcsin\lambda.$$

At this moment the energy equation gives $\dot{u}^2 = k^2[(\alpha - \beta)^2 - (\frac{\pi}{2} - \alpha)^2] > 0$, so the beads are still moving apart and carry on past the ends of the diameter.

Ask yourself What happens after the beads pass the ends of a diameter?

For $u \geq \frac{\pi}{2}$ we have $\theta = \pi - u$, so the energy equation reads

$$\dot{u}^2 = k^2[(\alpha - \beta)^2 - (u - (\pi - \alpha))^2],$$

which is SHM in u about $u = \pi - \alpha$, again with angular frequency k. This is the mirror image, in $u = \frac{\pi}{2}$, of the motion before the diameter. So the beads take another time t_1 to come to rest at $u = \pi - \beta$ (where $\theta = \beta$ again, with the beads now on the far side of O), then retrace the whole path back to $u = \beta$. The beads return to their starting positions after four such stages, so the period of the motion is

$$T = 4t_1 = \frac{4}{k}\left(\frac{\pi}{2} + \arcsin\frac{\frac{\pi}{2} - \alpha}{\alpha - \beta}\right) = \frac{2}{k}\left(\pi + 2\arcsin\frac{\pi - 2\alpha}{2(\alpha - \beta)}\right).$$

(The angle θ itself, which forgets which side of O the beads are on, repeats after $2t_1$.)

Case 3: $\beta = 2\alpha - \frac{\pi}{2}$

Now θ reaches $2\alpha - \beta = \frac{\pi}{2}$ at $t = \frac{\pi}{k}$ exactly when the SHM brings the beads to rest. They are then at opposite ends of a diameter, so the spring lies along AOB and its force on each bead is along the radius: it has no component along the hoop, and nothing else acts along the hoop. The beads are in equilibrium and at rest, so they stay there. So

$$\beta = 2\alpha - \frac{\pi}{2}$$

is the value for which no oscillations occur. This matches the two periods: as β approaches this value from below, $\lambda \to 1$ and the Case 2 period tends to $\frac{4\pi}{k}$, twice the Case 1 period, because the beads linger longer and longer near the diameter.

Checks

- Take $k = 1$, $\alpha = \frac{3\pi}{8}$, $\beta = \frac{\pi}{8}$, which is in Case 2 since $2\alpha - \frac{\pi}{2} = \frac{\pi}{4}$. Then $\lambda = \frac{\pi/8}{\pi/4} = \frac{1}{2}$, $\arcsin\lambda = \frac{\pi}{6}$, and $T = 2(\pi + \frac{\pi}{3}) = \frac{8\pi}{3} \approx 8.378$.

- We stepped the two beads' equations of motion numerically (with the energy as a function of both bead positions, and no symmetry assumed). With $k = 1$, $\alpha = 1.2$, $\beta = 0.5$ the beads returned to rest at their starting points after 8.5162, against the formula's 8.51622; with $\beta = 1.0$ (Case 1) the period was $6.2832 \approx 2\pi$.
- Dimensions: k has units of s^{-1} (the energy is ma^2k^2 times an angle squared), so $\frac{2\pi}{k}$ is a time.

Common slips

- Writing the kinetic energy as $\frac{1}{2}ma^2\dot{\theta}^2$: both beads move, so it is $ma^2\dot{\theta}^2$. Missing the factor of 2 gives the wrong angular frequency $\sqrt{2}\,k$.
- Assuming the beads move symmetrically without a reason: it follows from conservation of angular momentum about O, which needs the spring force on each bead to act along the same line AB.
- Forgetting that θ is acute, and so answering $\frac{2\pi}{k}$ in every case.

Beyond STEP

Reducing a two-particle system to one coordinate using a conserved quantity is the start of Lagrangian mechanics, where symmetries (here, rotating the whole configuration about O) give conservation laws, an idea made precise by Noether's theorem. The motion in Case 2 is an oscillation in a "double-well" potential, whose period depends on the amplitude and jumps where the energy equals the height of the central barrier. For more on conservation laws and oscillations, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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